Where, ω is angular velocity and α is angular acceleration. We can also write equation 5.26 as,

$$\tau = \frac{d(I\omega)}{dt}$$

$$\tau = \frac{dL}{dt} \qquad (5.27)$$

The above expression says that an external torque on a rigid body fixed to an axis produces rate of change of angular momentum in the body about that axis. This is the Newton's second law in rotational motion as it is in the form of $F = \frac{dp}{dt}$ which holds good for translational motion.

Conservation of angular momentum:

From the above expression we could conclude that in the absence of external torque, the angular momentum of the rigid body or system of particles is conserved.

$$\text{If } \tau = 0 \text{ then, } \frac{dL}{dt} = 0;\ L = \text{constant}$$

The above expression is known as law of conservation of angular momentum. We will learn about this law further in section 5.5.

5.3 EQUILIBRIUM OF RIGID BODIES

When a body is at rest without any motion on a table, we say that there is no force acting on the body. Actually it is wrong because, there is gravitational force acting on the body downward and also the normal force exerted by table on the body upward. These two forces cancel each other and thus there is no net force acting on the body. There is a lot of difference between the terms "no force" and "no net force" acting on a body. The same argument holds good for rotational conditions in terms of torque or moment of force.

A rigid body is said to be in mechanical equilibrium when both its linear momentum and angular momentum remain constant.

When the linear momentum remains constant, the net force acting on the body is zero.

$$\vec{F}_{net} = 0 \qquad (5.28)$$

In this condition, the body is said to be in translational equilibrium. This implies that the vector sum of different forces $\vec{F}_1, \vec{F}_2, \vec{F}_3 \ldots$ acting in different directions on the body is zero.

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \cdots + \vec{F}_n = 0 \qquad (5.29)$$

If the forces $\vec{F}_1, \vec{F}_2, \vec{F}_3 \ldots$ act in different directions on the body, we can resolve them into horizontal and vertical components and then take the resultant in the respective directions. In this case there will be horizontal as well as vertical equilibria possible.

Similarly, when the angular momentum remains constant, the net torque acting on the body is zero.

$$\vec{\tau}_{net} = 0 \qquad (5.30)$$

Under this condition, the body is said to be in rotational equilibrium. The vector sum of different torques $\vec{\tau}_1, \vec{\tau}_2, \vec{\tau}_3 \ldots$ producing different senses of rotation on the body is zero.

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